KENYATTA UNIVERSITY

EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE IN CIVIL ENGINEERING

ECV 206: SURVEYING II

1^{ST} SEMESTER 2021/2022 TIME: 2 Hours

INSTRUCTIONS:

Attempt questions ONE and any other TWO questions

QUESTION ONE (30 Marks)

a) Distinguish between the precision and reliability as used in measurement (2 Marks)

b) Describe the three types of traverses used to establish survey controls (3 Marks)

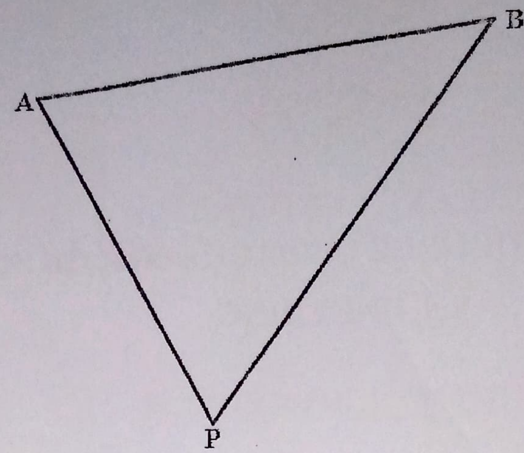
c) The following survey was carried out from the bottom of a shaft at A, along an existing tunnel to the bottom of a shaft at E. (5 marks)

Line	WCB °	′	″	Measured distance (m)	Remarks
AB	70	30	00	150.00	Rising 1 in 10
BC	0	00	00	200.50	level
CD	154	12	00	250.00	level
DE	90	00	00	400.56	Falling 1 in 30

i. If the two shafts are to be connected by a straight tunnel, calculate the bearing A to E and the grade.

ii. If a theodolite is set up at A and backsighted to B, what is the value of the clockwise angle to be turned off, to give the line of the new tunnel?

d) In the figure below, A and B have coordinates 1870.74mE, 953.56mN and 2013.53mE, 1020.23mN respectively. Determine the coordinates of P if, distance AP and BP have been measured by EDM as 143.94m and 178.08m respectively. (6 Marks)



e) With the aid of well labelled diagrams, describe the principle behind the stadia method in tacheometry *(10 Marks)*

f) A traverse *ACDB* was surveyed by theodolite and tape. The lengths and bearings of the lines *AC*, *CD* and *DB* are given below:

Line	*AC*	*CD*	*DB*
Length (m)	480.6	292	448.1
Bearing	25°19'	37°53'	301°00'

If the coordinates of *A* are x = 0, y = 0 and that of *B* are x = 0, y = 897.05, adjust the traverse and determine the coordinates of C and D. The coordinates of A and B must not be altered. *(4 Marks)*

QUESTION TWO **(20 Marks)**

a) Outline the methods used in the establishment of the survey controls *(4 Marks)*

b) Outline the computation steps in a traverse *(8 Marks)*

c) Explain how the coordinates of a point can be determined by observations to known points. *(8 Marks)*

QUESTION THREE **(20 Marks)**

a) Describe how a control point could be obtained using the intersection method *(7 Marks)*

b) A level railway is to be constructed from A to D in a straight line, passing through a large hill situated between A and D. In order to speed the work, the tunnel is to be driven from both sides of the hill.

The centre-line has been established from A to the foot of the hill at B where the tunnel will commence, and it is now required to establish the centre-line on the other side of the hill at C, from which the tunnel will be driven back towards B. To provide this data the following traverse was carried out around the hill:

Line	*WCB* °	′	″	*Horizontal distance (m)*	*Remarks*
AB	88	00	00	-	Centre line of railway
BE	46	30	00	495.8m	
EF	90	00	00	350.0m	
FG	174	12	00	-	Long sight past hill

Calculate: *(13 Marks)*

i. The horizontal distance from F along FG to establish point C.

ii. The clockwise angle turned off from CF to give the line of the reverse tunnel drivage.

iii. The horizontal length of tunnel to be driven.

QUESTION FOUR *(20 Marks)*

a) Describe the steps involved in setting up a theodolite *(7 Marks)*

b) Outline the sources of error in traversing *(3 Marks)*

c) The coordinates of A, B and C in the figure below are:

E_A 1234.96m N_A 17 594.48m

E_B 7994.42m N_B 24 343.45m

E_C 17 913.83m N_C 21 364.73m

And the observed angles are:

APB = α = 61° 41' 46.6''

BPC = β = 74° 14' 58.1''

1

2 Attach tribrach

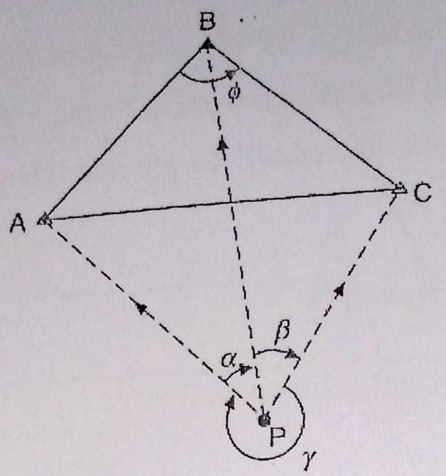
3 Centre optical plummet line of sight

4 Foot screws centre circular bubble

5 Centre plate bubble

6 Unclamp & rotate laterally

7 Repeat 4 5 & 6



Find the coordinates of P. *(10 Marks)*

QUESTION FIVE **(20 Marks)**

A link traverse below commences from known stations, *A* and *B*, and connects to known stations *C* and *D*. The bearing of the link AB is 151°27'38'' and that of CD is 347°37'41''. The observed angles and distances are as in the table below.

Station	Observed angle	Line	Distance (m)
B	143°54'47''	B- E_1	651.16
E_1	149°08'11''	E_1- E_2	870.92
E_2	224°07'32''	E_2- E_3	522.08
E_3	157°21'53''	E_3- E_4	1107.36
E_4	167°05'15''	E_4- C	794.35
C	74°32'48''		

Given that the coordinates of point B & C are (3854.28, 9372.98) & (7575.56, 8503.21) respectively, compute the adjusted coordinates of points E_1, E_2, E_3, E_4. *(20 marks)*

ECV 206 Surveying II CAT II

a) A new control station, F, is to be established from an existing coordinated points T and D (Fig. 1). The horizontal clockwise angles at T and D have been observed as DTF = 44° 52′ 36″ and TDF = 284° 26′ 38″ respectively. Calculate the coordinates of station F given that the coordinates of T are 3931.82 mE, 7491.98 mN and of D are 2959.39 mE, 7487.09 nN. *(5 marks)*

$$E_F = \frac{N_T - N_D + E_T \cot D + E_D \cot T}{\cot T + \cot D}$$

$$N_F = \frac{E_T - E_D + N_T \cot D + N_D \cot T}{\cot T + \cot D}$$

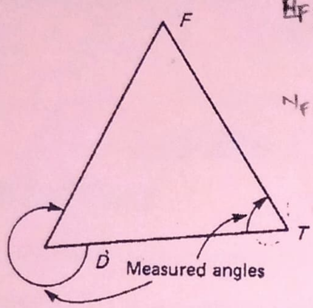


Figure 1

b) Angles have been observed at station F (Fig. 2) between the coordinated control points P, W and D so that the position of the station can be determined by resection.

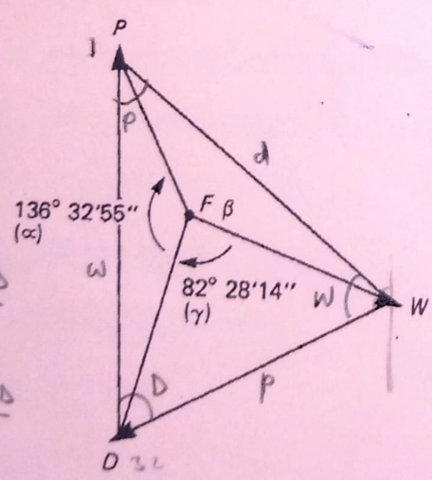


Figure 2

$$\frac{E_P + k_2 E_W + k_3 E_D}{k_1 + k_2 + k_3}$$

$$\frac{N_P + k_2 N_W + k_3 N_D}{k_1 + k_2 + k_3}$$

$$\frac{1}{\cot P - \cot \gamma}$$

$$\frac{1}{\cot W - \cot \alpha}$$

1

$$d = \sqrt{P^2 + W^2 + 2PW\cos D}$$

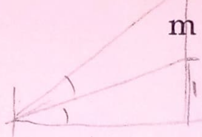
$$P = \sqrt{d^2 + W^2 - 2dW\cos P}$$

Calculate the coordinates of F, given the coordinates of P, W and D as follows: *(5 marks)*

P	2876.24 mE	8754.11 mN
W	3810.80 mE	7997.25 mN
D	2959.39 mE	7487.07 mN

c) Describe the basic principle of satellite position fixing *(6 marks)*

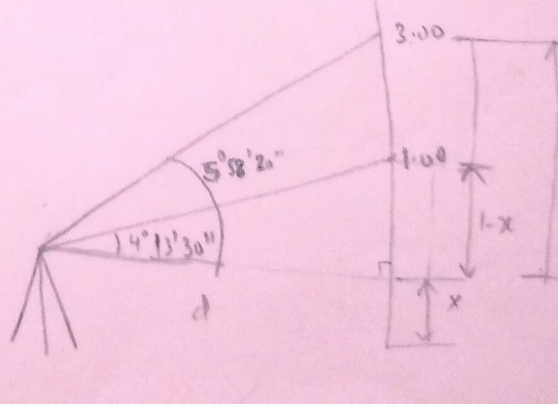
d) The following readings were taken on a vertical held staff. Calculate the horizontal distance between the theodolite and the staff, and the elevation of the staff station, if the height of the instrument axis is 37.36 m above datum. *(4 marks)*



Vertical angle	Staff reading
+4°13′30″	1.00
+5°58′20″	3.00

e) What do you understand by the terminology trigonometric heighting? *(2 marks)*

f) Derive an equation for the elevation difference between stations A and B taking into consideration the combined effects of curvature and refraction *(8 marks)*



13.53

$d = \frac{1+6}{0.07}$

$= 97.3$

ECV 206 Surveying II CAT I

a) Describe the three types of traverses used to establish survey controls *(3 marks)*

b) A theodolite traverse carried out between two points *A* and *B* on the line of a proposed tunnel resulted in the following observations being obtained:

Line	Bearing	Length (m)
A-(i)	086° 37′	128.88
(i)-(ii)	165° 18′	208.56
(ii)-(iii)	223° 15′	96.54
(iii)-(B)	159° 53′	145.05

Calculate the bearing and distance of point *B* from point *A*. *(4 marks)*

c) The following survey was carried out from the bottom of a shaft at A, along an existing tunnel to the bottom of a shaft at E. *(6 marks)*

Line	*WCB*			*Measured distance (m)*	*Remarks*
	o	*′*	*″*		
AB	70	30	00	150.00	Rising 1 in 10
BC	0	00	00	200.50	level
CD	154	12	00	250.00	level
DE	90	00	00	400.56	Falling 1 in 30

i. If the two shafts are to be connected by a straight tunnel, calculate the bearing A to E and the grade.

ii. If a theodolite is set up at A and backsighted to B, what is the value of the clockwise angle to be turned off, to give the line of the new tunnel?

d) Explain the basic features of a typical theodolite *(13 marks)*

e) What is the whole- circle bearing and the quadrantal bearing of a line joining point A (1023.25 mE, 2134.32 mN) to point C (1141.71 mE, 2012.89 mN)? *(2 marks)*

f) Given the coordinates of A as (EA, NA =1600.00, 5600.00), bearing and distance of A to B as 120° 50′24″@156.205 meters and compute the coordinates of B. *(2 marks)*

ECV 206 Surveying II CAT I

a) Describe the three types of traverses used to establish survey controls (3 Marks)

b) A theodolite traverse carried out between two points *A* and *B* on the line of a proposed tunnel resulted in the following observations being obtained:

Line	Bearing	Length (m)
A-(i)	086° 37′	128.88
(i)-(ii)	165° 18′	208.56
(ii)-(iii)	223° 15′	96.54
(iii)-(B)	159° 53′	145.05

Calculate the bearing and distance of point *B* from point *A*. (4 Marks)

c) Describe the meaning of the following terms when used in the context of a survey control traverse: (6 Marks)

i. Forward and back bearing

ii. Angular misclosure

iii. Bowditch adjustment

iv. Coordinate misclosure

d) Explain the basic features of a typical theodolite (13 marks)

e) What is the whole- circle bearing and the quadrantal bearing of a line joining point A (1023.25 mE, 2134.32 mN) to point C (1141.71 mE, 2012.89 mN)? (2 Marks)

f) Given the coordinates of A as (E_A, N_A =1600.00, 5600.00), bearing and distance of A to B as 120° 50′24″@156.205 meters and compute the coordinates of *B*. (2 marks)